

Scenario 2

Scenario 1: Personalized E-Commerce Recommendations

You are developing an e-commerce chatbot using ChatGPT API. The chatbot struggles to provide accurate product recommendations, and users are often dissatisfied.

Question:

How would you enhance the chatbot's ability to generate personalized recommendations?

Logical Solution:

- Implement RAG using a vector database like FAISS to retrieve product embeddings.
- Provide user-specific data (like purchase history or preferences) as context.
- Use ChatGPT API with the retrieved data to generate personalized suggestions.
- Adjust the model's temperature for balanced creativity and accuracy.

Scenario 2: Real-Time News Summarization

A media platform wants to build a real-time news summarizer using ChatGPT API. The challenge is ensuring the summaries are up-to-date and relevant.

Question:

How would you ensure the chatbot generates real-time summaries without relying solely on its pre-trained data?

Logical Solution:

- Use a RAG pipeline to fetch real-time news articles using external APIs.
- Convert the articles to embeddings and store them in a vector database.
- Retrieve the latest articles using similarity search.
- Provide the retrieved articles as context to the ChatGPT API for generating accurate summaries.

Scenario 3: Legal Assistance Chatbot

A law firm is using ChatGPT API to build a chatbot that provides legal advice. The chatbot sometimes generates speculative responses instead of fact-based answers.

Question:

What approach would you take to ensure the chatbot provides reliable legal information?

Logical Solution:

- Implement a RAG system to fetch relevant case laws and regulations from a legal document database.
- Use domain-specific embeddings for accurate document retrieval.

- Configure ChatGPT API to generate responses using the retrieved legal content as context.
- Enable human review for critical legal queries to ensure compliance.

Scenario 4: Customer Support Automation

A telecom company has deployed a customer support chatbot using ChatGPT API. Users report that the chatbot often provides inconsistent answers for similar queries.

Question:

How can you improve the chatbot's consistency and ensure reliable responses?

Logical Solution:

- Implement context management using conversation history to maintain continuity.
- Fine-tune the ChatGPT API on telecom-specific data to reduce ambiguity.
- Use RAG to retrieve relevant support documents or FAQs.

Adjust parameters like temperature and top_p for more deterministic responses.

Scenario 5: Healthcare Diagnostic Chatbot

A healthcare provider uses ChatGPT API to offer preliminary medical advice. However, the chatbot occasionally provides medically inaccurate information.

Question:

How would you ensure the chatbot offers accurate and safe medical guidance?

Logical Solution:

- Implement a RAG system that retrieves medical data from verified databases like PubMed or CDC.
- Use strict temperature control to minimize hallucinations.
- Add a validation layer that cross-checks generated responses with medical knowledge.
- Set up escalation mechanisms to route complex queries to healthcare professionals.